DANIEL LAZAREV

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Mathematician and physician developing mathematical, statistical, and machine learning methods for information-theoretic inference and optimal data representations, with applications to large biological datasets.

EDUCATION

Massachusetts Institute of Technology — PhD, Mathematics
Advisors: Bonnie Berger (MIT), Benjamin M. Neale (Broad Institute, HMS) — *2021 – 2026*
Thesis: "Entropy-Extremizing Representations"

Renaissance School of Medicine at Stony Brook University — MD
2018 – 2026

Yeshiva University — BA (Hons.), Mathematics, Physics
2012 – 2016

EMPLOYMENT AND RESEARCH EXPERIENCE

Department of Mathematics, MIT — Cambridge, MA
Graduate Researcher — *September 2021 – present*

- Developing a mathematical framework for optimal data representations based on entropy extremization principles, with the goal of characterizing representations that are optimal relative to specified structural or informational constraints.
- Studying nonlinear generalizations of independent component analysis using tools from optimal transport, functional analysis, probability theory, and information geometry.
- Developing novel entropy-based model selection, reproducibility, and model weight significance metrics for latent representation models, with applications to interpretable machine learning and statistical genetics.
- Developing generalized entropy-based methods with applications to representation learning, information theory, geometry, and statistical inference.

Computer Science and Artificial Intelligence Lab (CSAIL), MIT — Cambridge, MA
Graduate Researcher, Computation and Biology Group, CSAIL — *September 2021 – present*

- Formulated a mathematical framework modeling evolution as diffusion constrained to dynamically evolving subspaces of genotype space, providing a principled probabilistic description of evolutionary dynamics under biological constraints.
- Developed `DiffEvol`, which infers latent evolutionary constraints directly from observed trajectory data by inverting constrained diffusion dynamics, enabling data-driven characterization of adaptive structure in viral evolution.
- Applied the framework to SARS-CoV-2 and influenza genomic datasets to study how constraint geometry shapes evolutionary trajectories and variant emergence.
- Connected diffusion-based evolutionary modeling with broader problems in representation learning, latent structure discovery, and stochastic dynamics on high-dimensional manifolds.

Broad Institute of MIT and Harvard — Cambridge, MA

Graduate Researcher — *September 2021 – present*
Associate Computational Biologist — *March 2020 – September 2021*

Neale lab in the Stanley Center for Psychiatric Research at the Broad Institute of MIT and Harvard and in the Analytic and Translational Genetics Unit at Massachusetts General Hospital.

- Developed GUIDE, a variational representation-learning framework for discovering latent biological structure from large-scale GWAS summary statistics, enabling hypothesis-free identification of shared genetic programs across complex traits and diseases. GitHub: `https://github.com/daniel-lazarev/GUIDE`. Interactive browser: `https://guide-analysis.hail.is/`.
- Designed scalable statistical methods for learning maximally informative low-dimensional representations of genomic data from biobank-scale datasets (e.g., UK Biobank and FinnGen), integrating techniques from variational inference, optimization, information theory, and high-dimensional statistics.
- Built open-source computational tools and interactive analysis infrastructure for large-scale genomic representation learning and downstream biological interpretation (Hail-based deployment and visualization tools).
- Collaborated with researchers including Bonnie Berger, Benjamin M. Neale, Henry Cohn, Alicia Martin, and Adit Radhakrishnan on statistical genetics, multimodal learning, and representation learning methods for biological data.

Applied Mathematics & Statistics Department, Stony Brook University Stony Brook, NY
Student Researcher *December 2018 – December 2020*

Mathematical Physics and Fluid Dynamics: The Euler and Navier-Stokes equations model fluid flow and turbulence, but admit multiple solutions, even when solved numerically. We proved that the maximum entropy production principle is a necessary admissibility condition for the physically relevant solution to those equations. Advised by Prof. James Glimm, in collaboration with Prof. Gui-Qiang Chen (University of Oxford).

Physics Department, Yeshiva University New York, NY
Student Researcher *January 2015 – May 2018*

Atomic Force Microscopy: Built a mathematical model to find the size and charge of a ring sample *in vacuo* and in electrolytic environments given data typically provided by an atomic force microscope. Advised by Prof. Fredy Zypman.

Mathematics Department, Yeshiva University New York, NY
Student Researcher *September 2014 – May 2017*

- Network Science: Investigated small-world networks, with application to the spread of cancer-promoting behaviors on college campuses. Advised by Prof. Marian Gidea.
- Nonlinear Dynamics: Analyzed the motion of a charged particle in the magnetic field created by a circular wire, perturbed by a constant, external magnetic field as a model for the motion of charged particles in accelerators and other magnetic instruments. Advised by Prof. Marian Gidea.

PUBLICATIONS AND PREPRINTS

8. **D. Lazarev**, "Stokes' theorem as an entropy-extremizing duality." Preprint: arXiv:2509.16386. *Submitted.*
7. **D. Lazarev**, G. Chau, A. Bloemendal, B. Berger, C. Churchhouse, and B. M. Neale, "Entropy-minimizing representations uncover latent structure in complex genetic architectures." *Submitted.*
   Preprint: bioRxiv:2024.05.03.592285.
6. **D. Lazarev**, A. Sappington, G. Chau, R. Zhang, B. Berger, "Constrained Diffusion as a Paradigm for Evolution." In *Research in Computational Molecular Biology: 30th International Conference, RECOMB 2026. Under review for RECOMB Special Issue at Genome Research.*
5. Y. H. Jee, Y. He, W. Lu, Y. Shi, **D. Lazarev**, M. J. Daly, M. P. Reeve, A. R. Martin, "Dissecting pleiotropy: approaches to gain mechanistic insights into human disease." *Nature Reviews Genetics* (2025).
4. **D. Lazarev**, "Information measures for entropy and symmetry" (2023). Preprint: arXiv:2211.14857.
3. J. Glimm, **D. Lazarev**, and G.-Q. Chen, "Maximum entropy production as a necessary admissibility condition for the fluid Navier-Stokes and Euler equations," *SN Applied Sciences* **2**, 2160 (2020).

2. **D. Lazarev** and F. R. Zypman, "Charge and size of a ring in an electrolyte with atomic force microscopy," *Journal of Electrostatics* **87**, 243 (2017).

1. **D. Lazarev** and F.R. Zypman, "Determination of size and charge of rings by atomic force microscopy," *Journal of Electrostatics* **83**, 69 (2016).

SELECTED MANUSCRIPTS IN PREPARATION

- **D. Lazarev*** and J. Elenbaas*, "Mendelian randomization using latent biological programs."
- **D. Lazarev**, D. Maksumov, G. Chau, S. Zhang, A. Radhakrishnan, B. M. Neale, and B. Berger, "Model weight significance measures for latent representation models."
- J. Elenbaas*, **D. Lazarev***, G. Chau, C. Churchhouse, and B. M. Neale, "Disease mechanism discovery using mech-GUIDE."
- **D. Lazarev**, "On entropy and structure."
- **D. Lazarev**, "Solution of Rényi's problem on the characterization of entropy."
- **D. Lazarev**, "Model selection by entropy minimization."

INVITED TALKS

- MGH, Analytic and Translational Genetics Unit Meeting (6/2023)
- MGH, Analytic and Translational Genetics Unit Meeting (10/2023)
- Harvard School of Public Health, Statistical Genetics Meeting (4/2024)
- MGH, Analytic and Translational Genetics Unit Meeting (5/2025)
- Novo Nordisk Foundation Center for Genomic Mechanisms, Broad Institute (11/2025)
- Brandeis University, Mathematical Biology Seminar (4/2026)
- Research in Computational Molecular Biology (RECOMB), 30th International Conference (5/2026)
- MGH, Analytic and Translational Genetics Unit Meeting (6/2026)
- Cross-Disorder Working Group of the Psychiatric Genomics Consortium (9/2026)

TEACHING, SERVICE, AND LEADERSHIP EXPERIENCE

Massachusetts Institute of Technology — Cambridge, MA

- *Teaching Assistant* (January 2026 – May 2026) for GenAI for Biology course (18.S997 / 6.8710 / 6.8711 / 20.390 / 20.490 / HST.506).
- *DRP Mentor* (Dec - Jan, 2021 - 2024): Served as a graduate mentor to 1-2 undergraduate mathematics students in the Directed Reading Program (DRP) during Winter Sessions of 2021-2022, 2022-2023, and 2023-2024.
- *PRIMES Mentor* (January 2023 – September 2023): mentored a high school student in MIT's Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES) during the calendar year 2023.

YOY High School — Brookline, MA
Science Teacher — *September 2025 – June 2026*

Designed and taught a senior Modern Science course.

Kochvei HaShamayim — Boston, MA
Founder and CEO — *February 2021 – present*

Kochvei HaShamayim is a nonprofit organization championing the values of family, community, and education by supporting young Jewish couples, allowing them to minimize difficult compromises and to excel in both their family life and their early career or continued education. Visit Kochvei.org for more information.

Yeshiva University, Physics Department New York, NY
Adjunct Instructor *August 2016 – May 2018*

Courses taught: Introduction to Physics I Lab, Introduction to Physics II Lab, General Physics II Lab, General Physics III Lab, General Physics I Problem Seminar, General Physics II Problem Seminar.

TABC High School Teaneck, NJ
Physics Teacher *January 2018 – June 2018*

Taught two classes of eleventh grade physics.

YRSRH Middle and High School New York, NY
Math and Science Teacher *September 2016 – June 2018*

Courses taught: Science (sixth and seventh grades), Algebra I: Regents Prep (eighth and ninth grades), Chemistry: Regents Prep (eleventh grade), AP Physics (twelfth grade), SAT Math Review (twelfth grade).

Yeshiva University New York, NY

- *Student Course Assistant* (September 2015 – May 2016) for General Physics (Honors)
- *Honors Program Advisor* (November 2015 – May 2016): Advised first in the Honors Program regarding coursework, and helped them devise a four-year course of study
- *Peer and Private Tutor* (January 2014 – May 2016): Calculus, Physics, Chemistry and Writing
- *Member of Student Government* (August 2013 – May 2016): Sophomore Class President ('13–'14), Junior Class President ('14–'15), Senior Justice of the Student Court ('15–'16)
- *Student Ambassador* (September 2013 – May 2016): Gave tours, participated in panel discussions, and represented the Mathematics Department, the Physics Department, and the Honors Program
- *Mentorship Program Volunteer* (January 2013 – January 2016): Helped run science modules for elementary school students in underrepresented schools and for young patients in children's hospitals as a member of several initiatives, including Project START, CollegeEDge, the YU Literacy Program, and Project TEACH

AWARDS AND ACHIEVEMENTS

- **Sigma Xi, The Scientific Research Honor Society**, Full Membership, July 2025.
- **The Reviewers' Choice Award** for abstract in American Society of Human Genetics (ASHG) Annual Meeting, given to the top 10% of abstracts. August 2023.
- **OU Impact Accelerator Grant:** Selected as one of five nonprofits out of over seventy organizations to be part of Cohort IV of the OU Impact Accelerator, which includes extensive professional training and a $10,000 grant. May 2022.
- **MathWorks Fellowship:** Full tuition and stipend support for the first year of graduate studies at MIT, valued at $100,000.
- **Jay and Jeanie Schottenstein Honors Program:** Additional Honors Program course requirements; Honors Thesis; full scholarship for undergraduate studies at Yeshiva U.
- **Professor Arnold & Bertha Lowan Memorial Award for Excellence in Physics Research:** Awarded April 2016 with a cash award of $1200.
- **Dr. Ron and Cheryl Nagel Award for Excellence in Pre-Medical Studies:** Awarded May 2016.

- **The Lawrence P. Fischer Memorial Award** for "the best Hebrew paper on some aspect of Jewish History." Awarded May 2016 with a prize of $1500.
- **Imrei Shefer Writing Contest:** First place in a Yeshiva U. writing contest with a prize of $1500. Awarded January 2016.
- **Dean's List:** 2012 – 2016.